msvtx_el4hcal

MC vs data · overview · rank 79 / 98

rank 79 / 98 by endcap D_equal

used by: NN1 barrel, NN1 endcap

rank-identical columns merged into this one:
summedCaloClusters/el4hcal
MS1Vtx_l4hcal_raw
MS1Vtx_barrel_l4hcal
MS1Vtx_endcap_l4hcal

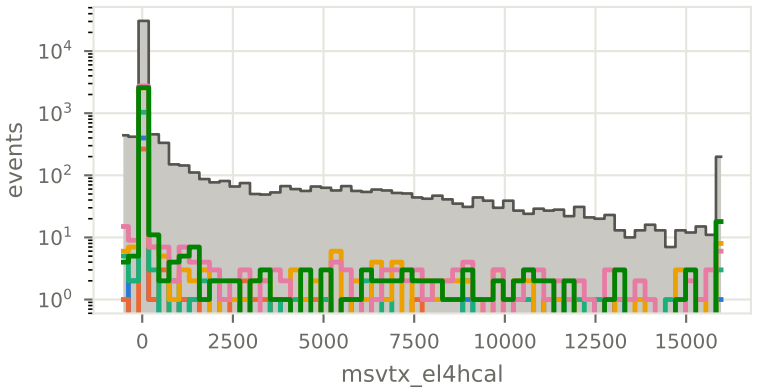
region	D_equal	noise	D_pooled	125/15	125/55	200/50	400/100	600/150	1000/275	n_bkg	rank
endcap	0.094	0.022	0.093	0.110	0.080	0.104	0.096	0.089	0.089	34,970	79/98
barrel	0.166	0.051	0.159	0.151	0.202	0.181	0.172	0.160	0.139	1,089	61/98
all	0.089	0.018	0.085	0.097	0.090	0.099	0.092	0.081	0.075	36,059	91/98

msvtx_el4hcal

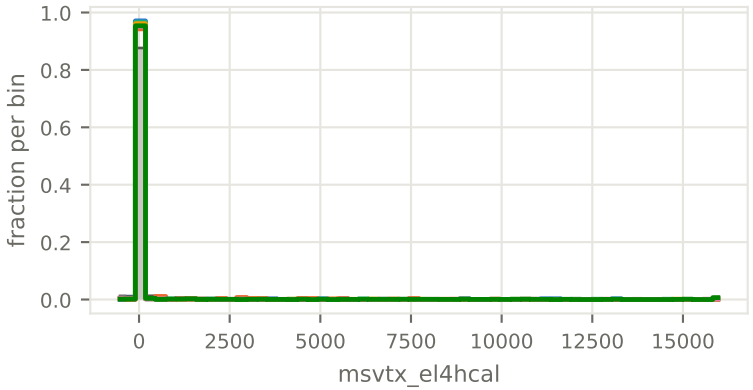
region: ENDCAP · rank 79 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.094 (noise 0.022) D_pooled 0.093 min 0.080 (mH125_mS55) max 0.110 (mH125_mS15) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

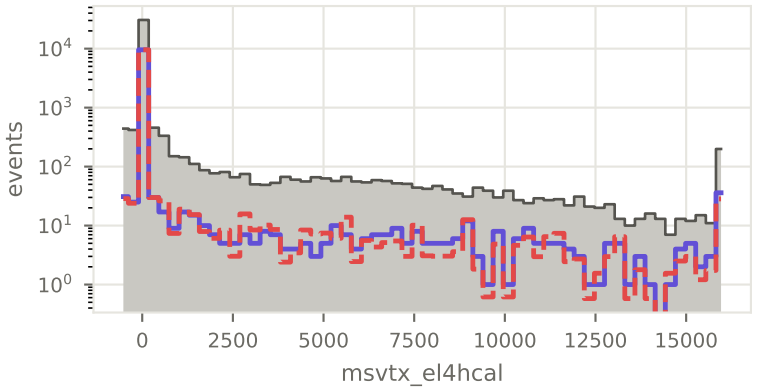


1b · each mass point vs background · normalized

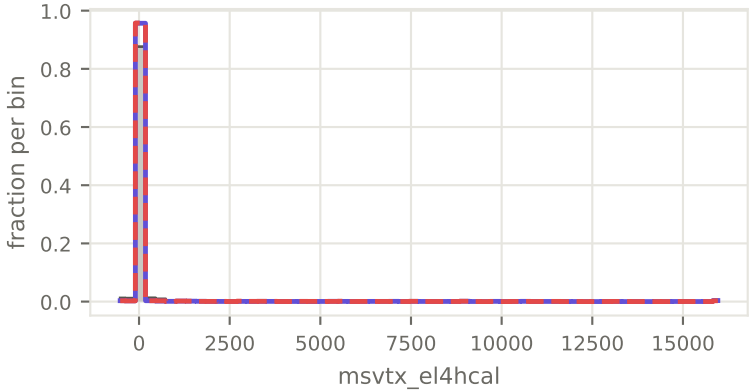


1 · each mass point
background n 34,970
mH125_mS15 D 0.110 n 411
mH125_mS55 D 0.080 n 281
mH200_mS50 D 0.104 n 1,071
mH400_mS100 D 0.096 n 2,648
mH600_mS150 D 0.089 n 2,882
mH1000_mS275 D 0.089 n 2,703

2a · combined signal vs background · raw

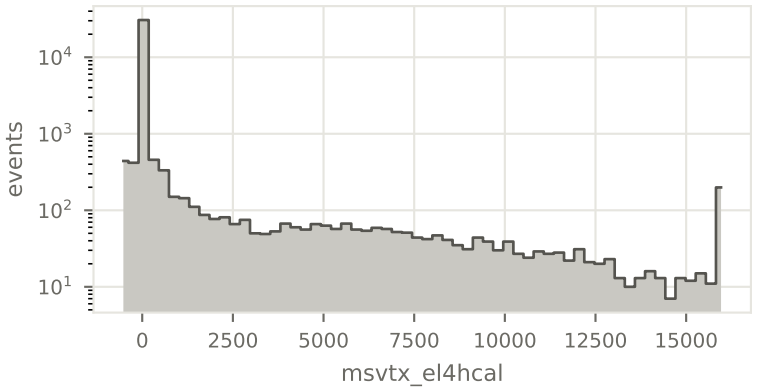


2b · combined signal vs background · normalized

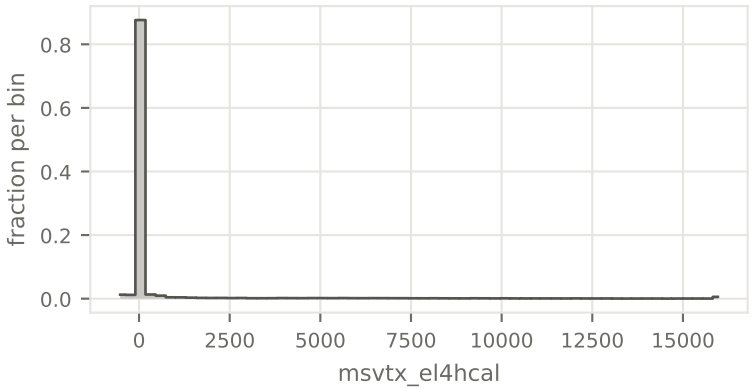


2 · combined signal
background n 34,970
combined, pooled D 0.093 n 9,996
equal weight per mass D 0.094

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

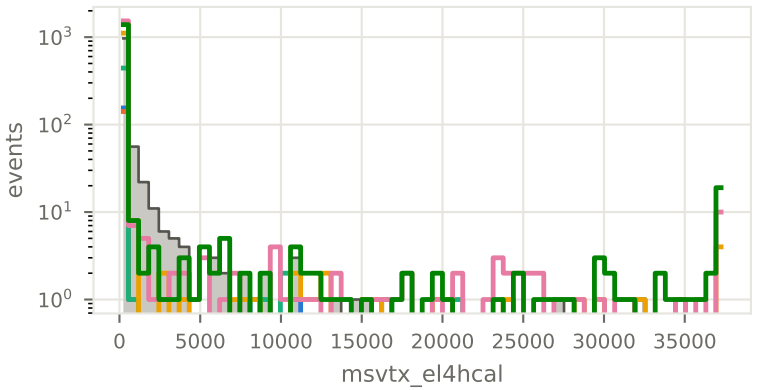
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_el4hcal

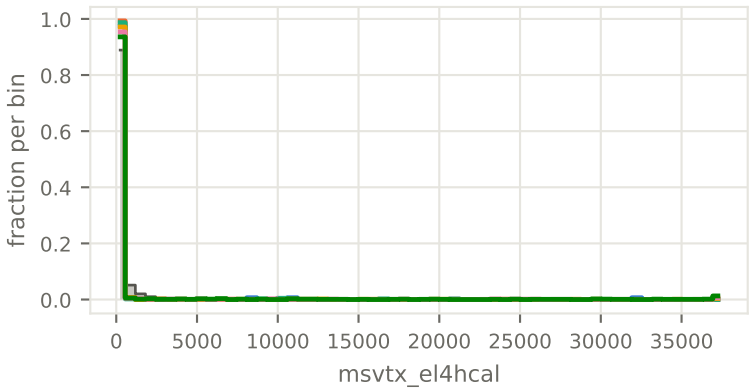
region: BARREL · rank 79 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.166 (noise 0.051) D_pooled 0.159 min 0.139 (mH1000_mS275) max 0.202 (mH125_mS55) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

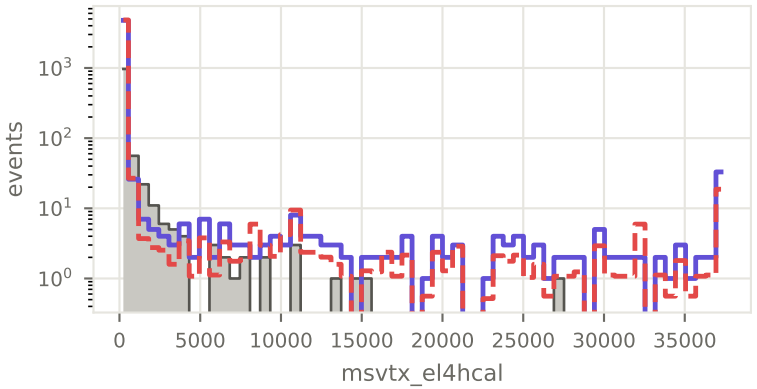


1b · each mass point vs background · normalized

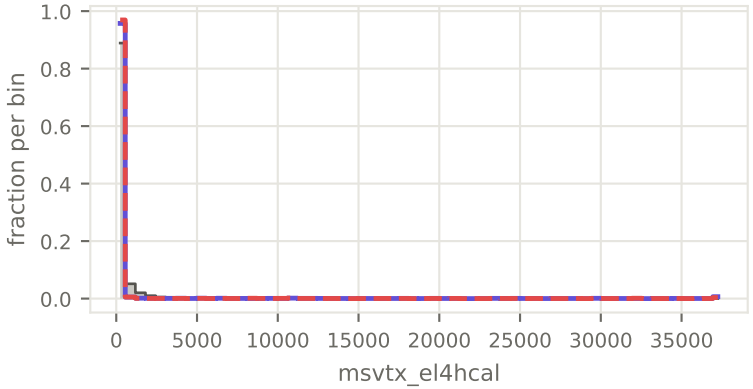


1 · each mass point
background n 1,089
mH125_mS15 D 0.151 n 159
mH125_mS55 D 0.202 n 142
mH200_mS50 D 0.181 n 449
mH400_mS100 D 0.172 n 1,144
mH600_mS150 D 0.160 n 1,608
mH1000_mS275 D 0.139 n 1,485

2a · combined signal vs background · raw

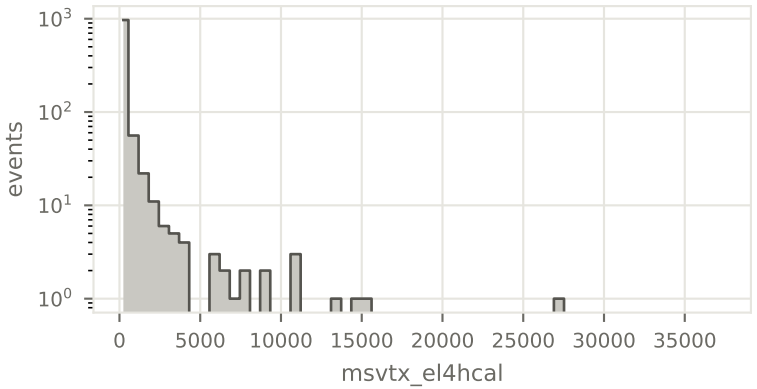


2b · combined signal vs background · normalized

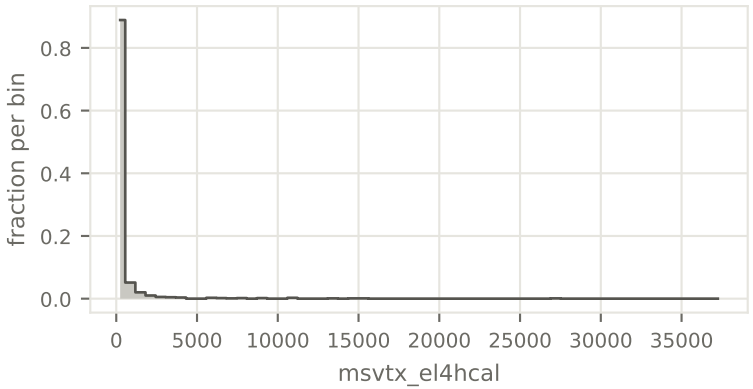


2 · combined signal
background n 1,089
combined, pooled D 0.159 n 4,987
equal weight per mass D 0.166

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

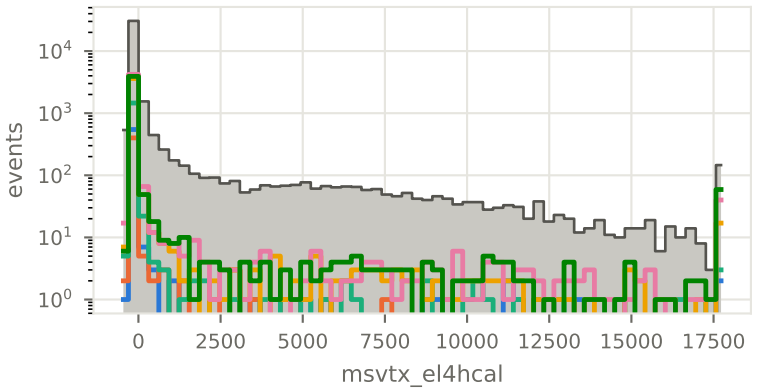
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_el4hcal

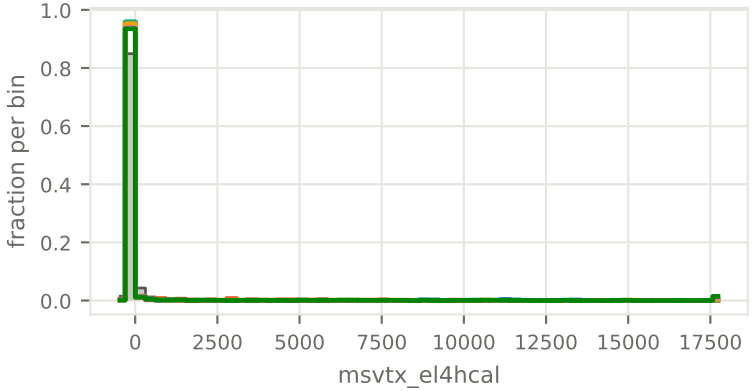
region: ALL · rank 79 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.089 (noise 0.018) D_pooled 0.085 min 0.075 (mH1000_mS275) max 0.099 (mH200_mS50) bkg n = 36,059 sig n = 14,983

1a · each mass point vs background · raw

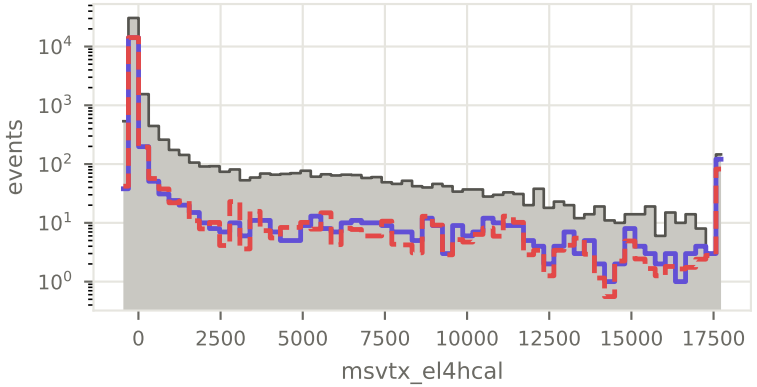


1b · each mass point vs background · normalized

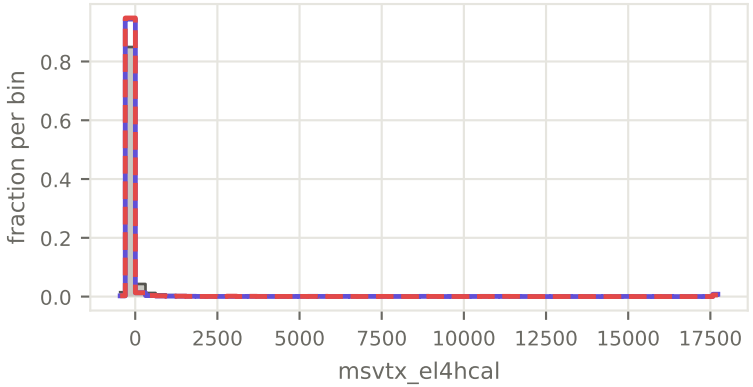


1 · each mass point
background n 36,059
mH125_mS15 D 0.097 n 570
mH125_mS55 D 0.090 n 423
mH200_mS50 D 0.099 n 1,520
mH400_mS100 D 0.092 n 3,792
mH600_mS150 D 0.081 n 4,490
mH1000_mS275 D 0.075 n 4,188

2a · combined signal vs background · raw

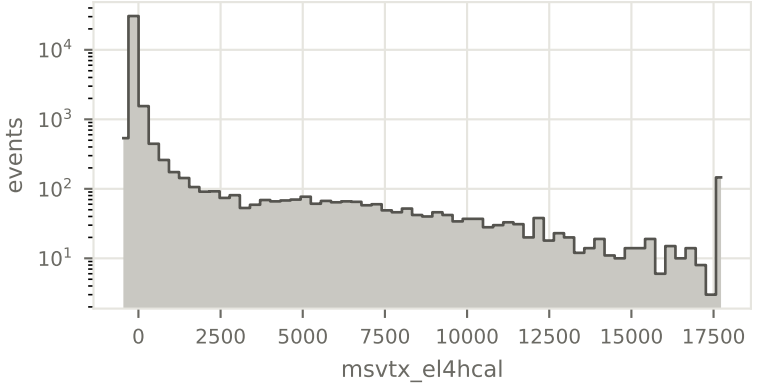


2b · combined signal vs background · normalized

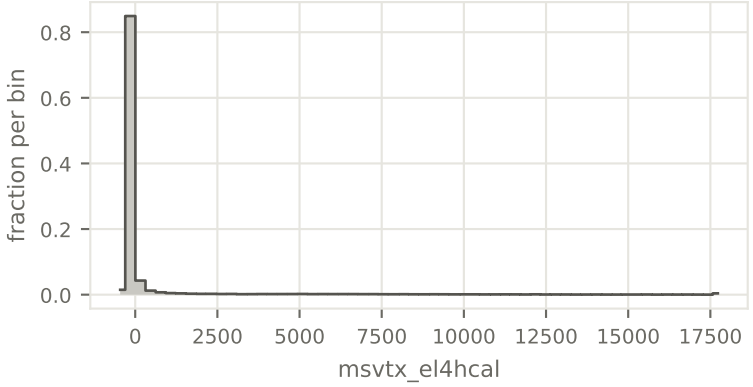


2 · combined signal
background n 36,059
combined, pooled D 0.085 n 14,983
equal weight per mass D 0.089

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 36,059

a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed